



AD23431 - STATISTICAL ANALYSIS AND COMPUTING**EXP 4: Functions and Strings in R**

1. Write an R function to clean a given text string by:

- converting all characters to lowercase,
- removing punctuation symbols,
- removing extra white spaces.

```
clean_text <- function(text) {  
  text <- tolower(text)  
  text <- gsub("[:punct:]", "", text)  
  text <- gsub("\\s+", " ", text)  
  trimws(text)  
}
```

2. Write an R function that takes a sentence as input and returns the total number of words present in the sentence.

```
count_words <- function(sentence) {  
  words <- unlist(strsplit(trimws(sentence), "\\s+"))  
  length(words)  
}
```

3. Write an R function to extract the username part from a given email ID.

```
extract_username <- function(email) {  
  strsplit(email, "@")[[1]][1]  
}
```

4. Write an R function to validate a phone number such that:

- it contains exactly 10 digits, and
 - it consists only of numeric characters.
- The function should return TRUE or FALSE.

```
validate_phone <- function(phone) {  
  grepl("^[0-9]{10}$", phone)  
}
```



5. Write an R function to replace missing textual values in a character vector.

Replace empty strings ("") and "NA" with the value "Unknown".

```
replace_missing <- function(text_vector) {  
  text_vector[text_vector == "" | text_vector == "NA"] <- "Unknown"  
  text_vector  
}
```

6. Write an R function that extracts all hashtags from a given sentence and returns them as a character vector.

```
extract_hashtags <- function(sentence) {  
  unlist(regmatches(sentence, gregexpr("#\w+", sentence)))  
}
```

7. Write an R function to convert a given name string into Title Case (first letter of each word capitalized).

```
title_case <- function(name) {  
  words <- strsplit(tolower(name), "\s+")[1]  
  words <- paste0(toupper(substring(words, 1, 1)), substring(words, 2))  
  paste(words, collapse = " ")  
}
```

8. Write an R function that checks whether a given message contains spam-related words such as "free", "win", or "offer". The function should return TRUE if spam words are detected, otherwise FALSE.

```
detect_spam <- function(message) {  
  spam_words <- c("free", "win", "offer")  
  message <- tolower(message)  
  any(sapply(spam_words, function(word) grepl(word, message)))  
}
```



9. Write an R function that takes a full name as input and returns the initials of the name in the format X.Y.Z.

```
get_initials <- function(full_name) {  
  words <- strsplit(full_name, "\\s+")[[1]]  
  paste(toupper(substring(words, 1, 1)), collapse = ".")  
}
```

10. Write an R function that accepts a sentence as input and returns a word frequency table showing the count of each word in the sentence.

```
word_frequency <- function(sentence) {  
  sentence <- tolower(sentence)  
  sentence <- gsub("[:punct:]", "", sentence)  
  words <- unlist(strsplit(sentence, "\\s+"))  
  table(words)  
}
```

Learning Outcomes:

Evaluation Procedure	Marks Awarded
Program(5)	
Execution(5)	
Viva(5)	
Total(15)	
Faculty Signature	

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